AJAY KUMAR

Full Stack Developer | Approximately 2.5 years of project experience

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PROFESSIONAL PROFILE

Full Stack Developer with approximately 2.5 years of project experience across medical-device documentation, molecular design and biotechnology websites. Works across React interfaces, Node.js and Python APIs, databases and asynchronous jobs, with knowledge of Docker deployment and ML service integration.

TECHNICAL SKILLS

Languages and web: JavaScript, Python, TypeScript, HTML and CSS.

Frontend: React, Redux Toolkit, React Router, Tailwind CSS, Vite, Axios, Framer Motion; molecular interfaces using Ketcher, Mol*, Three.js and ReactFlow.

Backend: FastAPI, Node.js, Express, REST APIs, SQLModel/SQLAlchemy, JWT sessions, Socket.IO and asynchronous workers.

Data and messaging: PostgreSQL, Apache Cassandra, Redis, Apache Kafka and S3-compatible object storage; prior knowledge of MongoDB and Firebase.

Deployment and collaboration: Git, GitLab, GitHub, Docker, Docker Compose, Nginx, Linux and Windows.

Machine learning: ML service integration, asynchronous model-job orchestration and machine learning fundamentals.

EMPLOYMENT DETAILS

[Employer name] | [Official job title] | [Employment dates]

PROJECT EXPERIENCE

MDR Technical Documentation and Approval Platform

Full stack contributor | Two-person team, both full stack | [Project dates]

Application for preparing medical-device technical documents, managing attachments, routing submissions through review and approval, and exporting PDF and Word files.

Stack: React, Redux Toolkit, Tailwind CSS, Node.js, Express, Apache Cassandra, Redis/Kue, Socket.IO, PDFKit, DOCX and Docker Compose.

- Contributed across frontend interfaces and backend APIs for structured submissions, supporting-file uploads, dashboards and document status tracking.
- Worked on a five-tier approval workflow with designation-based access, approval/rejection actions and reviewer routing.
- Contributed to field-level edit history recording earlier values, updated values and the editing user, alongside deadline assignment and scheduled reminders.
- Worked on queued PDF/Word generation with retries and document regeneration after edits, together with user and role-based Socket.IO notifications.
- Worked with a Docker Compose environment combining frontend, backend, Cassandra and Redis services with persistent database volumes.

EDUCATION

Chandigarh University — **2021-2023**
Master of Computer Applications — *Punjab*

Armapore PG College — **2018-2021**
Bachelor of Arts — *Uttar Pradesh*

PROJECT EXPERIENCE CONTINUED

NuMol Molecular Design Platform

Frontend and backend contributor | Backend team: 4 | Frontend team: 2 | [Project dates]

Full stack scientific application covering molecule generation and optimization, docking, ADMET property prediction and retrosynthesis, with interactive frontend workflows connected to backend APIs and computational services.

Stack: React, Redux Toolkit, Vite, Tailwind CSS, Python, FastAPI, PostgreSQL, SQLModel/SQLAlchemy, Kafka, S3-compatible storage, Docker and Nginx.

- Contributed across frontend and backend in collaboration with the four-person backend team and two-person frontend team.
- Worked on API-connected molecular interfaces, including Ketcher structure editing, Mol*/Three.js protein and binding-pocket views, and sortable, paginated results.
- Contributed to asynchronous Kafka workflows that submit jobs to ML services, poll progress, persist results and handle cancellation, heartbeat leases and restart recovery.
- Worked on database-backed scheduling and computational-job accounting, including concurrency controls, token deductions, transaction records and refunds.
- Contributed to cookie-based JWT sessions, token refresh and CSRF handling across frontend and backend, and to scientific result workflows such as ADMET views and retrosynthesis graphs.
- Worked with Docker configurations separating API, worker and infrastructure services, plus a multi-stage frontend build served through Nginx with HTTPS configuration.

Micro CRISPR Website

Sole developer | [Project dates]

Independently developed a biotechnology company website using React, JavaScript, Vite, Tailwind CSS, Framer Motion, Lenis, EmailJS and Google Analytics 4.

- Built seven views with reusable components, responsive navigation, animated transitions and a scroll-driven video hero.
- Created a filterable therapeutic pipeline covering six modalities, with development-stage views, selectable programs and mobile layouts.
- Integrated career applications with resume attachments, input validation, asynchronous EmailJS submission and success/error handling; added GA4 page-view and navigation tracking.
- Implemented keyboard focus styles, semantic controls and reduced-motion support alongside responsive CSS.

PERSONAL PROJECT

Ligand Explorer - Ligand Interaction Diagram **[Project dates]**

Full stack application for analyzing uploaded protein-ligand structures and generating interactive two-dimensional SVG contact diagrams.

Stack: React, JavaScript, Redux Toolkit, Vite, Python, FastAPI, RDKit, PLIP, MDAnalysis, Open Babel, SVG, Docker Compose and Nginx.

- Developed structure-upload and ligand-discovery APIs with ligand/pose selection, configurable pocket radius and optional SDF input for ligand chemistry.
- Integrated PLIP detection for eight contact classes and RDKit molecular preparation, with atom mapping and weighted contact-score breakdowns.
- Implemented SVG residue placement, label collision handling and contact-edge routing; added contact selection, visibility filters and interaction inspection.
- Containerized frontend and backend with health checks; isolated PLIP execution with temporary directories, timeout, retry and concurrency controls.

DRAFT - Complete the marked dates and employment details, and confirm individual shared-project contributions before applying.